DC300 Operations Manual

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Record of Amendments

The information in this manual will be subject to revision in accordance with alterations made to the equipment specification or descriptive text.

The edition will be raised when the product specification is changed, and the issue will be raised when the descriptive text is changed.

Users of the manual should ensure that they are in possession of the correct edition and issue for the equipment in their possession. This information will be provided on the list of technical documentation supplied when the equipment is dispatched.

Manual	Issue	Date	Details
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1 GENERAL INFORMATION

1.1 Introduction

The DC-300 is a display console for monitoring the speed, torque, power, and temperatures available on a dynamometer. Its ergonomic design promotes portability and readability when placed on a desktop. The DC-300 features a touchscreen LCD display readable from any angle for simple visualization, calibration, and configuration. Channel data can be logged to a microSD card for later analysis and plotting, or it can be sent directly to other screens and devices using the host link options. The serial interface over USB-C is useful for grabbing DC-300 data from third-party Windows based DAQ systems, and the BLE wireless interface allows users to see dynamometer data from their mobile phones or tablets.

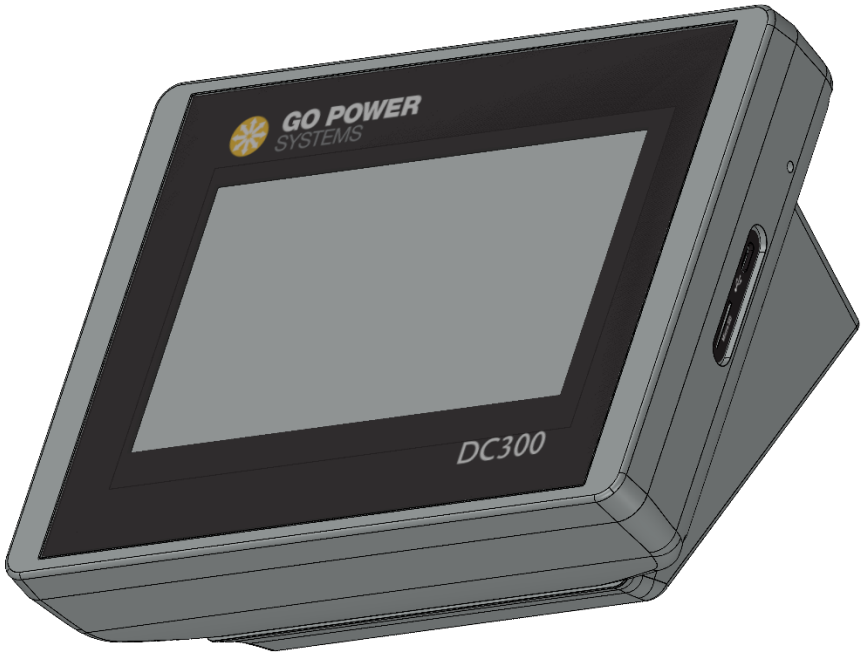

Figure 1 - DC-300 General View

1.2 Equipment Specification

Power:	*Supply:*	12-24VDC, 10W
Dimensions:	*Dimensions (mm):*	165 H x 225 W x 118.5 D
	Tilt angle:	20°
	Weight:	TBC
Environment:	*Temperature:*	-20 to 70°C (0 to 160°F)
	Humidity:	0 to 90% RH
Display:	*LCD:*	7” IPS
	Touch Type:	Resistive
	Viewing area (mm):	88.52 H x 156.68 W
Torque:	*Excitation:*	5VDC
	Input:	±7 mV to ±5 V (Bidirectional)
	Gain:	1 to 128
	Resolution:	System accuracy is dependent on the dynamometer. <0.25% Full Load for Froude dynamometers <1% Full Load for GoPower dynamometers
	Features:	50 / 60 Hz rejection
Tach:	*Interface:*	Inductive pulse pick-up
	AC Input:	40mV to 300V p-p 25kHz max
	Resolution:	± 1 RPM
Thermocouples:	*Connector:*	Mini Type-K
	Range:	-200°C to +700°C
	Resolution:	±2°C
	Features:	Open Thermocouple Detection
Logging:	*Removable media:*	MicroSD and MicroSDHC (up to 32GB)
Host Interfaces:	*Wired:*	Serial over USB-C
	Wireless:	Bluetooth BLE
Looms:	*TBC*	Tach / Torque Mil-Spec splitter (Optional for GoPower dynamometers)
	TBC	USB-C to USB-A serial cable
	TBC	Mini K-Type thermocouple M-F
Software:	*Serial Client:*	Froude DCX Client
General:	*Max log rate:*	100Hz (10ms)

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2 INSTALLATION

2.1 Dynamometer and Power Connections

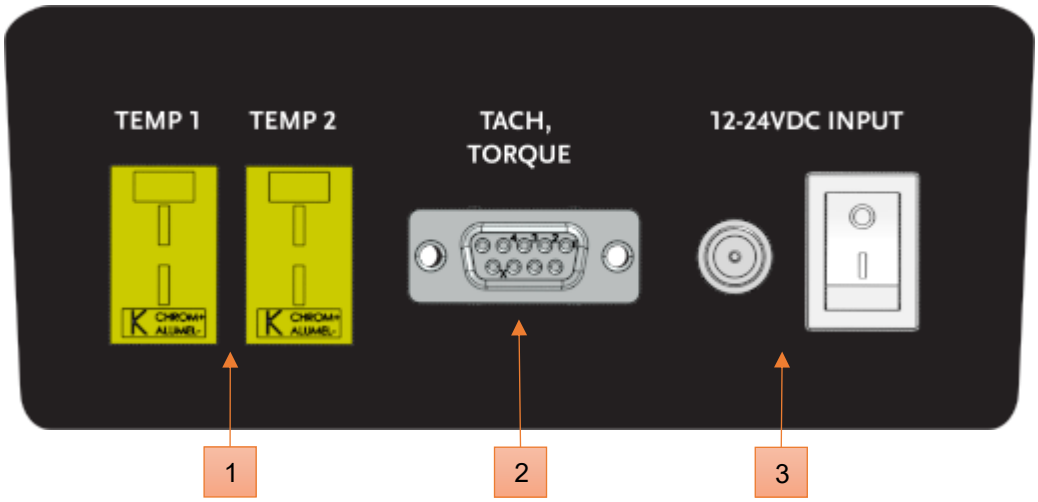

Figure 2 – Rear Side View

At the rear of the DC-300 are connectors for interfacing to the dynamometer. 2 K-Type thermocouple plugs **[1]** are provided for reading the inlet and outlet water temperature on hydraulic type dynamometers (GoPower series). As all channel names are customizable, these thermocouple inputs could also be repurposed for bearing temperatures on Eddy / AC dynamometers instead.

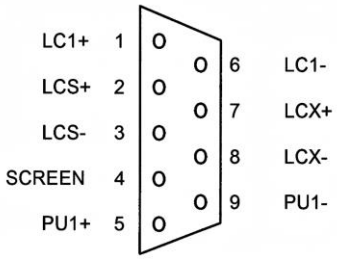

Figure 3 - Tach Torque Pinout

A female DB-9 connector **[2]** is used to connect the dynamometer tach and torque.

PU1+ and PU1- wire to a passive inductive pulse pickup tach sensor.

LCX+ and LCX- provide 5VDC excitation to a loadcell bridge. The feedback is read on pins LC1+ and LC1- from which the torque is determined. LCS+ and LCS- are sense wires: they are used to help compensate for any voltage drop on the cable. If the load cell has only 4 wires, LCS+ and LCS- should be tied to LCX+ and LCX- at the load cell respectfully.

The DC-300 is powered from the barrel power connector **[3]**. A power cable is supplied with the DC-300 that converts local mains supply (100-240VAC) to the 12-24VDC needed by the DC-300. A small rocker switch allows the console to be switched on and off without needing to remove the power cable. The DC-300 can safely be powered off at any time (there is no pre-defined shutdown sequence).

2.2 Logging Connections

Figure 4 - Right Side View

The ability to pull information from the DC-300 to be read on another computer / device is made possible using the microSD card slot **[1]**, the USB-C slot **[2]**, or the wireless interface (see section 4 for more information). The microSD and USB-C ports are accessible on the right side of the DC-300.

The microSD card is inserted into the DC-300 with the contacts facing upwards. A microSD adaptor may be required to fit the microSD to the SD card slot on many PCs. Note that MicroSDXC cards are not compatible with the DC-300.

The USB-C port on the DC-300 utilizes the data lines of the USB-C cable for serial diagnostics and configuration. Be sure not to use a charging only USB-C cable between the DC-300 and the host computer.

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3 OPERATION

3.1 Power On

Figure 5 - Splash Screen

When the DC-300 power cable is connected, and the ON/OFF switch at the rear of the unit is set to '1', the DC-300 will power on.
The green LED to the right side of the DC-300 will illuminate, and the LCD will display a splash screen for several seconds (Figure 5).
The splash screen indicates the current product version information. Any resources will be loaded at this time (like the GoPower logo), and any startup tasks will be executed.

3.2 Navigation

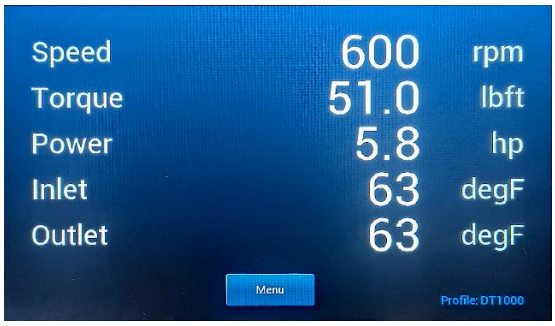

Figure 6 - Main View

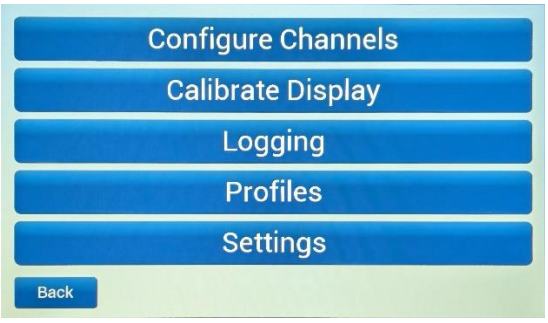

Figure 7 - Menu View

The DC-300 features a 7" touchscreen display for user visualization and customization.

The Main View (Figure 6) is shown by default on the DC-300. It shows all DC-300 channel names, values, and units. The presentation of each Channel is customizable (see section 0). The active profile name is also indicated in the bottom right for reference.

The Menu View (Figure 7) is accessed by touching the "Menu" button on the Main View. All user options can be accessed from this view as defined below. The "Back" button can be touched on any view to return to the proceeding view.

3.2.1 Editing Values

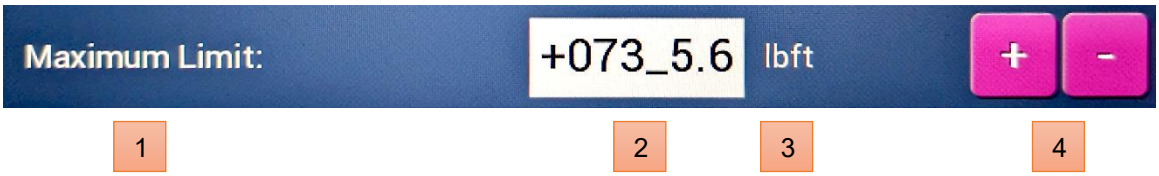

Figure 8 - Numeric Editing

Numeric fields like "Log Rate" or "Max Limit" can be changed by the user. An example of an editable numeric field is shown in Figure 8 where the field name **[1]**, value **[2]**, unit **[3]**, and adjustment buttons **[4]** can be found.

For simple whole numbers like “Number of Decimal Places”, there are only a few possible options, so the + and – buttons simply increase or decrease the value respectively. The change to the value is automatically applied.

For numbers that are much bigger, incrementing or decrementing by 1 could take a long time, and so a different approach is used. Touching either the value field **[2]**, or the adjustment buttons **[4]** will trigger editing of the value, and a flashing underscore position indicator will appear on the digit being affected. For example, Figure 8 shows the value 7345.6 being edited… the digit “4” is flashing. The +/- adjustment buttons are used to increment or decrement only the flashing digit, so if the + button was touched, the edited value would now read 73**5**5.6. To edit the digit left of the current editing digit, touch the value field. A touch to any other part of the screen will exit editing mode, and any changes will be applied.

3.2.2 Editing Text

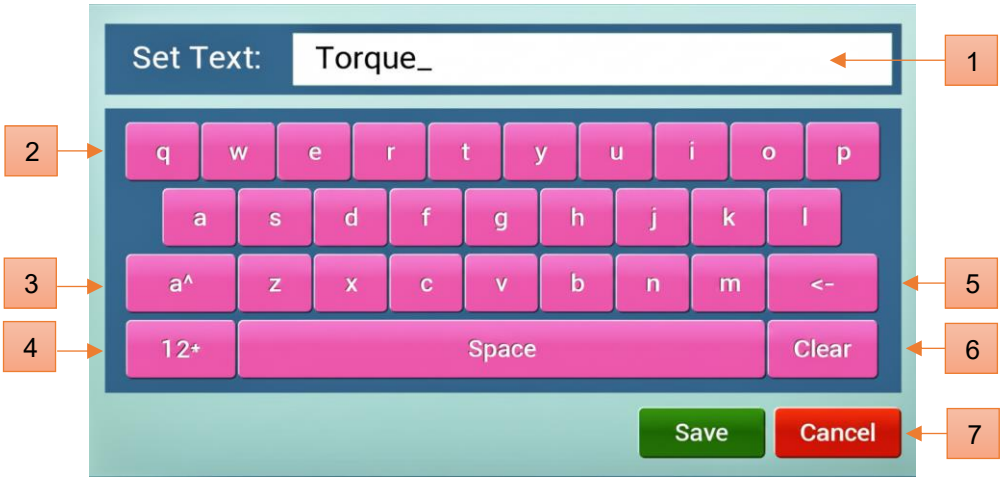

Figure 9 - Keypad View

Text fields like “Channel name” or “log file name” can be changed by the user. When these text-fields are touched, the keypad editor will show (Figure 9). The current text being edited **[1]** appears with a flashing position indicator. Standard alpha-numeric characters **[2]** can be appended by touching them. Uppercase characters can be accessed using the *a^* button **[3]**, and numeric and special characters can be accessed using the *12** button **[4]**. The <- button **[5]** can be used to remove only the last character before the position indicator, and the *clear* button **[6]** can be used to clear all text in the “Set Text” box.

The *Save* and *Cancel* buttons **[7]** will exit the editor where *Save* will commit any text changes and *Cancel* will discard them.

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3.3 Channel Configuration

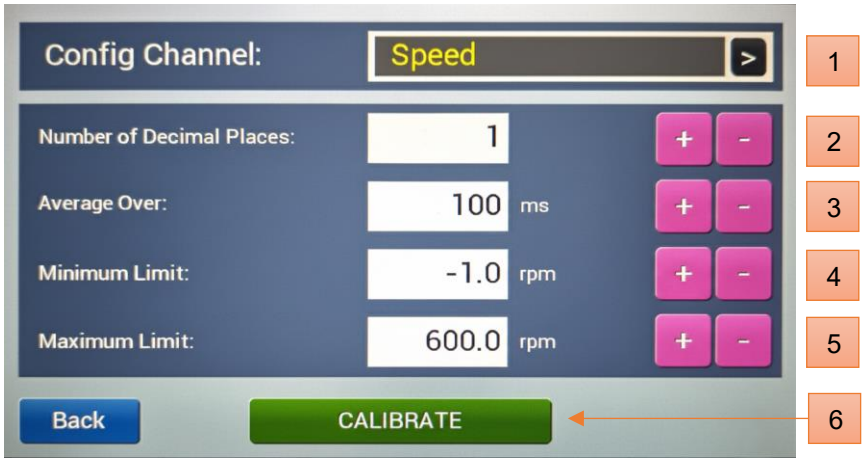

Figure 10 - Channel Configuration

The “Channel Configuration” view (Figure 10) can be accessed by touching “Configure Channels” in the Menu view. In this view, all attributes of a channel can be changed to meet the user requirements.

The header **[1]** shows the current channel being configured. The Channel name (yellow) can be touched, and its text edited. The next button (“>”) will switch the active channel to the next in the channel list- all parameters below will update to reflect the new channel selection.

The presentation of the channel value on the Main view can be adjusted by setting the number of decimal places **[2]** and the averaging time **[3]**. The averaging time (milliseconds) indicates how fast the on-screen value updates- a value of 0ms will present the channel value as soon as it is available. Longer averaging times are less prone to noise as all samples collected during the waiting period are averaged. A default time of 100ms has been chosen to allow the screen value to update fast enough to present the channel data with high real-time accuracy, but slow enough that the digits don’t flicker.

Each channel can be alarmed. If the channel value drops below the Minimum limit **[4]**, or rises above the Maximum limit **[5]**, then the channel value will flash orange in the Main View (Figure 11). This can be used as an indicator for the operator to abort the test.

Figure 11 - Alarm Triggered

The active channel can be calibrated by touching the “Calibrate” button **[6]**. This option will be greyed out for channels that do not support custom calibration.

3.4 Channel Calibration

The “Channel Calibration” view can be accessed by touching “Calibrate” in the “Channel Configuration” view.

Calibration is the method by which a system converts a raw signal (like voltage or frequency) into a useful “engineering value” (like torque or temperature). The process involves applying a known torque / temperature / etc. and informing the system what quantity has been added. The system can then associate with whatever raw value it currently sees (this is known as a calibration point). 2 or more calibration points are needed for a system to be able to present any real-world value. A maximum of 3 calibration points is currently supported.

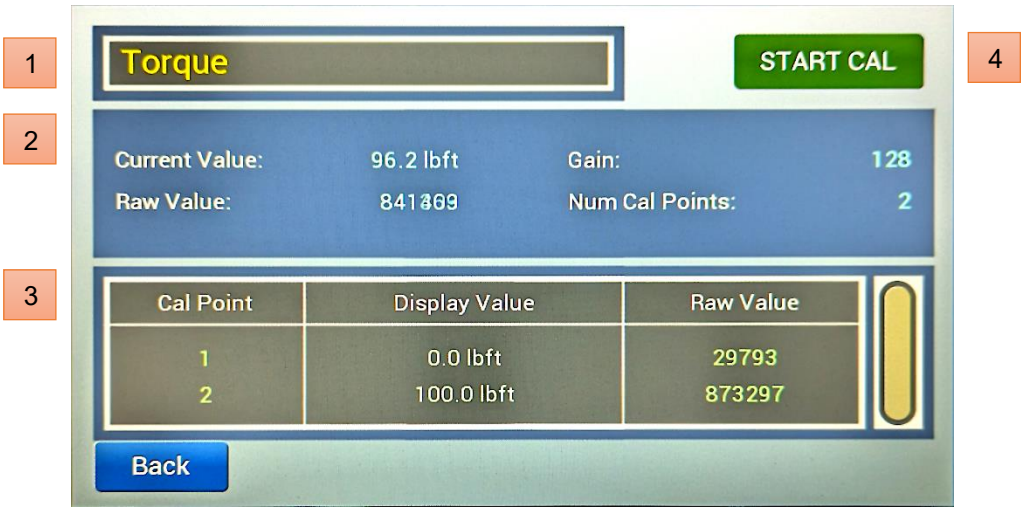

Figure 12 - Channel Calibration Overview

The “Channel Calibration” view appears as in Figure 12: The header indicates the channel whose calibration data is being shown **[1]** – a new calibration can be started by touching the “Start Cal” button **[4]**. Current calibration information about the channel is shown in a table indicating the stored calibration points **[3]** and a groupbox showing the current engineering value, raw value, gain and number of calibration points used in the last calibration **[2]**.

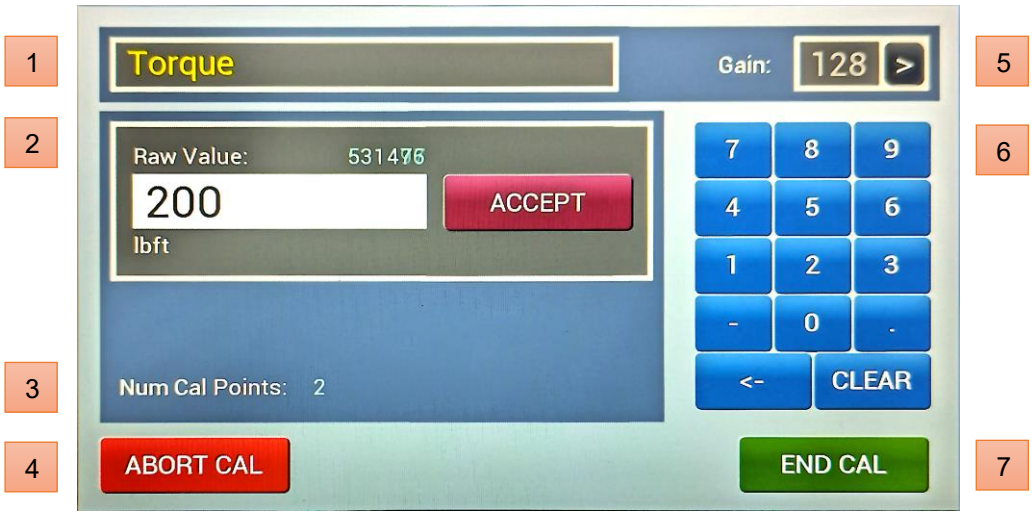

Figure 13 - Channel Calibration Execution

Upon touching the “Start Cal” button, the Calibration procedure begins as shown in Figure 13. The calibration can be aborted at any time by touching the “Abort Cal” button **[4]** – this will ignore any calibration points set and restore the previous channel calibration.

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Initially, the number of calibration points **[3]** will be 0. A standard linear calibration will have 2 calibrations points (a zero and a span); if calibrating a temperature channel for example, having the calibration points be 0°C and 100°C would allow the system to work out when 50°C (or any other temperature) is being seen by linear extrapolation.

Beneath the channel name **[1]** is the active calibration point **[2]**. The raw value is the digital representation of what the system can see- this value should change as the torque / temperature / etc. being calibrated changes. Be sure that the raw value can change, and that it stabilizes before accepting any calibration point.

The engineering value in the white edit box should match whatever physical quantity is being applied (e.g. In Figure 13, 200lbft of load should be applied). The engineering value can be changed using the numeric keypad **[6]**.

Some signals (like torque) can be very weak, and so an amplifier is needed to boost the signal. The Gain field **[5]** shows how much the signal is being amplified. If this option is available for the calibrated channel, the next button (>) is used to change the gain. As the gain increases, the raw value should increase also. A higher gain will mean higher resolution and accuracy, but too high a gain could result in signal saturation (where large temperatures / torque / etc. all appear the same). It is recommended to use the largest gain available, but to check that the raw value changes to cover your entire testing range. The gain field (if configurable) will only be editable when setting the first calibration point.

On setting the engineering value and matching the real-world quantity, the calibration point can be stored by touching the “Accept” button. The number of calibration points will increase to indicate the calibration point was accepted. When 2 or more calibration points have been defined, the “End Cal” button **[7]** will appear- touching this button will close the calibration procedure view and apply the new calibration points.

A typical Calibration procedure:

1) Touch the “Start Cal” button in the “Channel Calibration Overview” window (Figure 12)
2) Adjust the external quantity (torque / temperature / etc.) and make sure the raw value changes
3) If the gain is adjustable, choose a large gain. Apply the maximum external quantity- if the raw value saturates, reduce the gain until it doesn’t
4) Set the external quantity to 0 and ensure the engineering value reads 0. Wait for the raw value to stabilize, and touch “Accept”. Ensure the number of calibration points increases from 0 to 1
5) Set the external quantity to the max expected under test conditions and update the engineering value to match. Wait for the raw value to stabilize, and touch “Accept”. Ensure the number of calibration points increases from 1 to 2
6) Touch “End Cal” to complete the 2-point linear calibration

3.4.1 Torque Calibration

Calibration of the torque measurement system is the most important factor in the accuracy and repeatability of the dynamometer. Load cells are not all exactly the same, and they are affected by age and time in use. Torque calibration should be a regular part of the operational routine and must be performed any time a load cell is replaced.

The calibration is performed by placing known weights on the end of an arm mounted on the dynamometer. The distance from the center of rotation of the dynamometer (on a level plane) to the point on the arm where weight is applied is known as the “effective arm length”. The actual torque as seen by the console is a sum of the torque applied by the weights and of the arm itself.

To determine the torque applied by the weights, use the following formulas:

$$Torque\ (Nm) = Mass\ (kg) \times 9.81 \times Effective\ Arm\ Length\ (m)$$

$$Torque\ (lbft) = Weight\ (lb) \times Effective\ Arm\ Length\ (ft)$$

If the dynamometer has a counter balancing arm, then the effective torque of the calibration arm will be zero. For GoPower dynamometers however, the torque applied by the arm will need to be determined using the following method if it is not already known:

1) Remove arm and all weights from the dynamometer
2) Accept the first calibration point as 0 Nm / 0 lbft
3) Add the arm and maximum number of weights to the dynamometer
4) Accept the second calibration point as the torque applied by the weights alone and end the calibration
5) Remove all weights leaving only the calibration arm attached to the dynamometer and record the displayed torque shown on the DC-300: This will be the effective torque applied by the calibration arm

Following a load-cell calibration, it is recommended to load and unload the dynamometer with weights to check for sticking and confirm linearity. A calibration sheet has been provided in APPENDIX A.

3.5 Touch Screen Calibration

If the touch screen buttons don't correctly respond to a user touch, the touch screen and display may need to have their co-ordinates synched. This can be achieved by recalibrating the display.

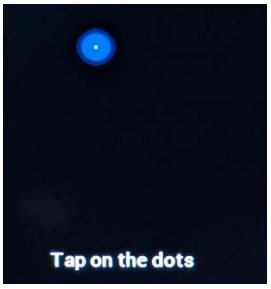

Figure 14 - Calibrate Display

The "Calibrate Display" view can be accessed by touching "Calibrate Display" in the Menu view, or by sending the serial command "TFT" (Section 4) (a better solution if the touchscreen is not responding correctly).

A pulsing blue dot with the caption "Tap on the dots" will show (Figure 14). The coordinate will be stored when the dot is touched- a total of 3 dots will need to be touched for the calibration process to succeed and the "Calibrate Display" view to close.

The touchscreen should respond correctly following this procedure.

3.6 Logging

All channels displayed on the DC-300 can be logged to a microSD card inserted into the right side of the unit. The exported file has a CSV extension and can be read natively by Microsoft Excel or any other text editor.

The "Logging" view can be accessed by touching "Logging" in the Menu view.

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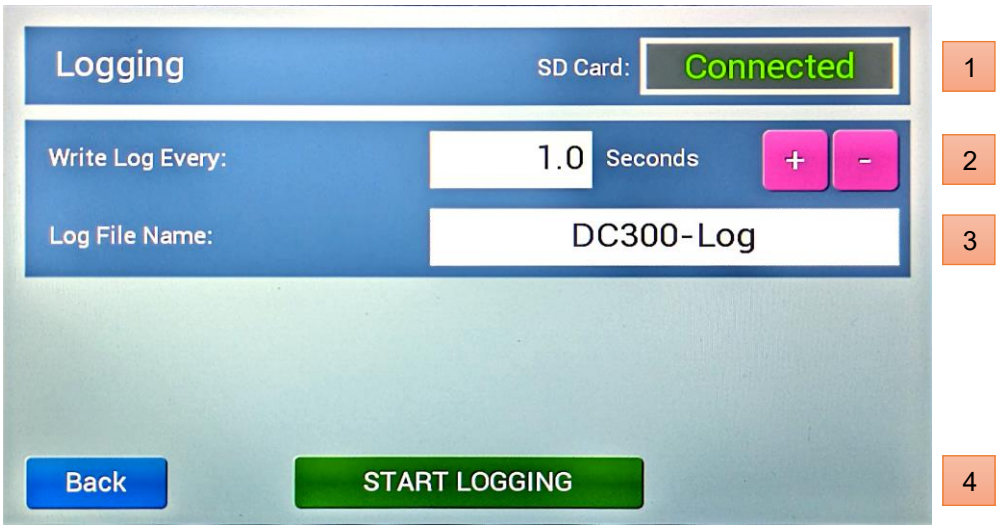

Figure 15 - Logging View

The status text **[1]** will indicate if the MicroSD card is plugged in and connected. A default name for the log file is provided **[3]**, however, when touched, a keypad will show, and the operator can assign a custom log name instead. If a log file already exists with the same name, a numeric suffix will be added to ensure the filename is unique. The file extension will be added automatically and does not need to be entered here. The log rate **[2]** determines how often a row will be added to the log file.

A log is started by touching the “Start Logging” button **[4]**. If the logging process starts successfully, this button will change to “Stop Logging” (Figure 16), and a flashing green indicator will appear on the Main View (Figure 17) to indicate log data is actively being written.

Figure 16 - Stop Logging

Figure 17 - Is Logging Indicator

A log can be stopped at any time by touching the “Stop Logging” button. Removing the MicroSD card during a log will also halt the logging but may result in corruption to the log file.

3.7 Profiles

All the custom channel names, limits, calibrations, and other configurations are stored in a profile. There are currently 3 custom profiles available on the DC-300 that can be switched between easily. This means that if more than one dynamometer is being used, each dynamometer could have its own profile with its own torque calibration. The advantage is that when the active dynamometer is swapped out, so too is the active profile, and there is no need to perform a new torque calibration.

Figure 18 - Profile View

The “Profiles” view can be accessed by touching “Profiles” in the Menu view.

Use the navigation buttons **[1]** to switch between profiles- the profile number and name **[2]** of the selected profile will be shown. Touching the Profile Name field will allow the name to be changed using the keypad view. To activate the selected profile, touch the “Activate” button **[3]** – all calibrations and channel configuration information from the profile will be automatically applied. If the selected profile is active, the button will show as “Activated” (Figure 19). The currently active profile is identified in the bottom right of the Main View.

Figure 19 - Activated Profile Button

3.8 Settings

Additional general settings can be accessed by touching “Settings” in the Menu view.

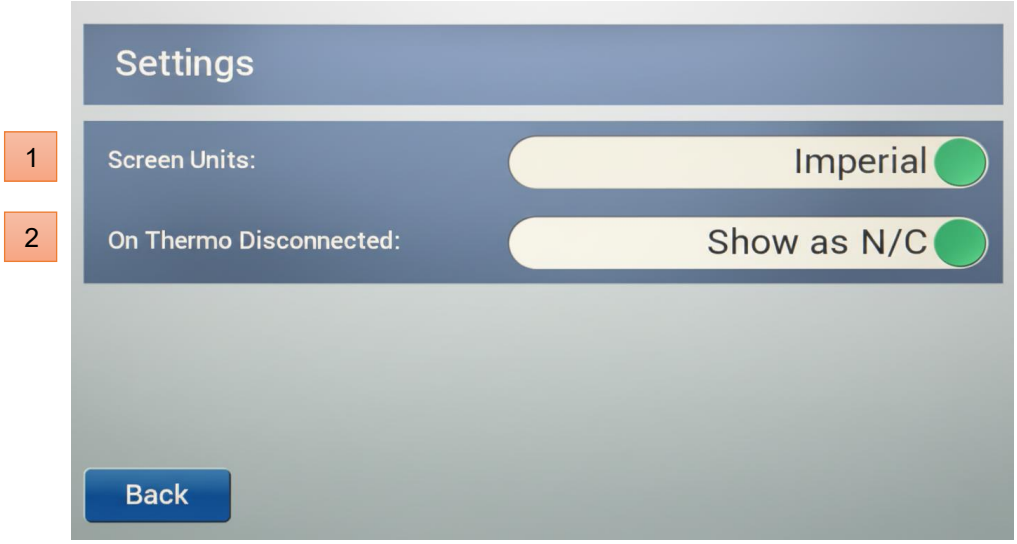

Figure 20 - Settings View

A toggle switch for “Screen Units” **[1]** allows the user to select between “Imperial” and “Metric”.

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When set to Imperial, the units on the Main View will show as degrees Fahrenheit (°F) for temperatures, pounds feed (lb-ft) for torque, and Horsepower (hp) for power.

When set to Metric, the units on the Main View will show as degrees Centigrade (°C) for temperatures, Newton meters (Nm) for torque, and Kilowatts (kW) for power.

A toggle switch for “On Thermo Disconnected” **[2]** allows the user to select what value is displayed if the temperature measurement detects there is a thermocouple fault.

When set to “Show as N/C”, a temperature channel value will indicate the error detected where “N/C” is an open circuit (thermocouple not connected), “VCC” indicates a short on the power line, “GND” indicates a short on the ground line, and “ERR” is any other detected error.

When set to “Do Nothing”, the temperature channel value will show whatever reading it detects- if there is a thermocouple error, this could be a large non-sensical number.

4 HOST INTERFACE

4.1 Serial Host

The DC-300 can be connected to a Windows computer with a USB-C cable. Windows should automatically recognize the DC-300 and install the correct driver.

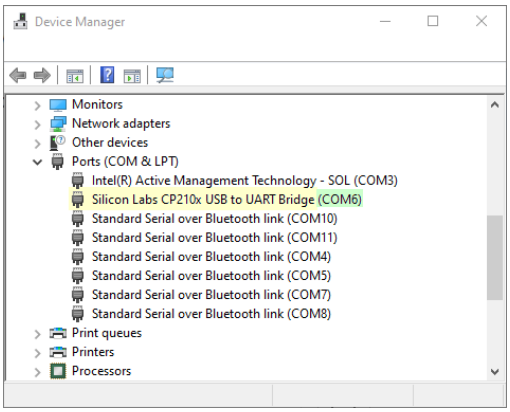

Figure 21 - Device Manager COM Port

In *Device Manager*, under "Ports", the DC-300 should show as "Silicon Labs CP210x USB to UART Bridge" as highlighted in yellow in Figure 21.

If Windows cannot recognize the DC-300, and it shows up as an "Unknown Device", install the latest serial driver here:

https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers

Take note of the COM port (highlighted green).

(The COM port number can be changed by right clicking the entry in *Device Manager,* and navigating to "Properties" -> Port Settings -> Advanced)

The DC-300 comes supplied with a Froude program called "DCX Client" (Section 4.1.2) for communicating and logging data from the DC-300 over the USB-C connection, however, any terminal application (e.g. Putty, TeraTerm, or Hyperterminal) can also be used.

The correct port settings needed for communicating with the DC-300 are shown in Figure 22 (where the port name should match the COM port noted in *Device Manager* above).

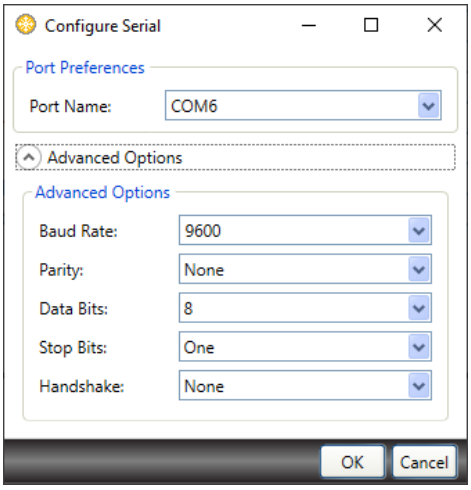

Figure 22 - DC-300 Serial Port Settings

4.1.1 Message Format

Messages sent and received are all human readable alphanumeric strings. The DC-300 acts as a server and will issue responses to commands requested by the client PC whilst also broadcasting data routinely when requested.

All messages are terminated by a carriage return and linefeed, and all elements in a message are separated by a space character.

The host command message format is:

<Command ID><…><CR><LF>

Where **<Command ID>** is a case insensitive 3-character command identifier, and **<…>** can be any number of command parameters separated by a space character.

Any issued command will be responded to with a message:

<Command ID><OK><CR><LF>

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If the message was received and processed successfully, or the message:

<Command ID><NOK><CR><LF>

If the message was received and could not be processed successfully.

DC-300 Commands and response messages are defined below.

4.1.1.1 Command: Help (HLP)

The “Help” command will return a list of all the supported commands from the DC-300 that can be requested by the host client.

Client must send **<Command ID>** as ‘**HLP**’ (or ‘**?**’) with no parameters.

For each DC-300 command supported, the DC-300 will respond with the line:

[?] <Ref Command ID> : <Ref Command Desc><CR><LF>

Where **<Ref Command ID>** is the Identifier, and **<Ref Command Desc>** is a text description of the Command being referenced.

```
HLP
HLP OK
[?] HLP : Help
[?] CAL : Calibration
[?] HDR : Channel Names
[?] UNT : Channel Units
[?] ENG : Channel Engineering Values
[?] RAW : Channel Raw Values
```

Figure 23 – Command ‘HLP’ Example

4.1.1.2 Command: Reset (RST)

The “Reset” command will restart the DC-300.

Client must send **<Command ID>** as ‘**RST**’ with no parameters.

No custom response will be returned.

4.1.1.3 Command: Calibrate Display (TFT)

The “Calibrate Display” command will trigger the touch screen calibration procedure (Section 3.5).

Client must send **<Command ID>** as ‘**TFT**’ with no parameters.

No custom response will be returned.

4.1.1.4 Command: Version Info (VER)

The “Version Info” command will return the DC-300 version number- this can be useful for diagnosing issues with the unit. The same version number is shown on the DC-300 splash screen.

Client must send **<Command ID>** as ‘**VER**’ with no parameters.

The DC-300 will respond with the line:

[V] <Version><CR><LF>

```
VER
VER OK
[V] DC300 V1.9
```

Figure 24 - Command 'VER' Example

4.1.1.5 Command: Channel Names (HDR)

The “Channel Names” command will return the names for each channel- these will be the custom names set by the user under the active profile.

Client must send **<Command ID>** as ‘**HDR**’ with no parameters.

The DC-300 will respond with the line:

[#]<TAB><Name 1> … <Name X><CR><LF>

Where each name is separated by a *tab* character.

```
HDR
HDR OK
[#]	Time	Speed	Torque	Power	Inlet	Outlet
```

Figure 25 - Command 'HDR' Example

4.1.1.6 Command: Channel Units (UNT)

The “Channel Units” command will return the units of each channel. Units will be imperial or metric based on the client request. Order of units matches names in HDR command.

Client must send **<Command ID>** as ‘**UNT**’ with no parameters.

The DC-300 will respond with the line:

[#]<TAB><Unit 1> … <Unit X><CR><LF>

Where each unit is separated by a *tab* character.

```
UNT
UNT OK
[U]	secs	rpm	lbft	hp	degF	degF
```

Figure 26 - Command 'UNT' Example

4.1.1.7 Command: Channel Engineering Values (ENG)

The “Channel Engineering Values” command will return the values of each channel. Units will be imperial or metric based on the client request. Order of values matches names in HDR command.

Client must send **<Command ID>** as ‘**ENG**’ with the following parameter:

<Tx Rate Ms> *Optional – integer*

The DC-300 will continually broadcast the “Channel Engineering Values” message at the rate defined in this parameter (milliseconds) without the client needing to resend the command. A value of 0 will disable automated broadcasting of this message.

If not provided, the DC-300 will send a single response only.

The DC-300 will respond with the line:

[M *or* I]<TAB><Val 1> … <Val X><CR><LF>

Where each value is separated by a *tab* character. **[M]** indicates the following values are given with metric units, where **[I]** indicates imperial units.

```
ENG
ENG OK
[M]	3809.700299.67	149.94	4.70	24.00	23.75
```

Figure 27 - Command 'ENG' Example

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4.1.1.8 Command: Channel Raw Values (RAW)

The “Channel Raw Values” command will return the raw uncalibrated values of each channel. This command is used for diagnostic purposes only. Order of values matches names in HDR command.

Client must send **<Command ID>** as ‘**RAW**’ with the following parameter:

<Tx Rate Ms> *Optional – integer*

The DC-300 will continually broadcast the “Channel Raw Values” message at the rate defined in this parameter (milliseconds) without the client needing to resend the command. A value of 0 will disable automated broadcasting of this message.

If not provided, the DC-300 will send a single response only.

The DC-300 will respond with the line:

[R]<TAB><Val 1> … <Val X><CR><LF>

Where each value is separated by a *tab* character.

```
RAW
RAW OK
[R]     4386    299     575170  0       24      23
```

Figure 28 - Command 'RAW' Example

4.1.1.9 Command: Number of Decimals (DEC)

The “Number of Decimals” command specifies to what decimal place accuracy the “Channel Engineering Values” should be returned to.

Client must send **<Command ID>** as ‘**DEC**’ with the following parameter:

<Num Dec> *Mandatory – integer*

The number of decimal places of values presented in the “Channel Engineering Values” message is defaulted to 1. Set this parameter to adjust.

No custom response will be returned, however subsequent calls to “Channel Engineering Values” will show the adjusted number of decimal places.

4.1.1.10 Command: Set Metric (MET)

The “Set Metric” command forces all channel values to be sent with metric units.

Client must send **<Command ID>** as ‘**MET**’ with no parameters.

Successful execution of this command will trigger the DC-300 to send the “Channel Names” and “Channel Units” packets (with the units correctly adjusted).

4.1.1.11 Command: Set Imperial (IMP)

The “Set Imperial” command forces all channel values to be sent with imperial units.

Client must send **<Command ID>** as ‘**IMP**’ with no parameters.

Successful execution of this command will trigger the DC-300 to send the “Channel Names” and “Channel Units” packets (with the units correctly adjusted).

4.1.1.12 Command: Calibrate Channel (CAL)

The “Calibrate Channel” command allows channels (like torque) to be calibrated from the host PC without needing to touch the DC-300 display.

Client must send **<Command ID>** as '**CAL**' with the following parameters:

<Channel ID> *Mandatory – integer*

The channel which is being calibrated or whose calibration information is being requested. Valid options are:

1 – *Time*

2 – *Speed*

3 – *Torque*

4 – *Power*

5 – *Temp 1 (Defaulted as "Inlet Temp")*

6 – *Temp 2 (Defaulted as "Outlet Temp")*

<Cal Action> *Conditional – text (case insensitive)*

Not required if reading calibration information only.

Defines the calibration action to be performed. Valid options are:

"START" – *Begin calibration*

"END" – *End calibration and save changes*

"ABORT" – *End calibration and discard changes*

"ACCEPT" – *Store this calibration point*

"CLEAR" – *Delete all calibration points associated with this channel*

<Eng Value> *Conditional – float*

Not required unless **<Cal Action>** is "ACCEPT"

Specifies the actual weight / torque, etc. being currently applied for this calibration point.

To any calibration command, the DC-300 will respond with the ***Calibration Status*** line:

[C0] <Chan Name>:<RUN or **CAL>**

Where **<Chan Name>** is the name of the channel whose calibration information is being referenced. The field following the colon will be **<CAL>** if the requested channel is currently being calibrated or **<RUN>** otherwise.

If applicable, the DC-300 will also send subsequent lines with any ***Calibration Points*** stored for the referenced channel with the format:

[C#] <Eng Value> <Eng Unit> = <Raw Value>

Where **#** is the calibration point number (e.g. '2' is the second calibration point), **<Eng Value>** and **<Eng Unit>** are the engineering value and unit of the current calibration point respectfully, and **<Raw Value>** is the raw value as seen by the DC-300 for this calibration point.

Read Channel Calibration

To read the calibration points stored for a channel, the sequence of messages sent and received is illustrated in Figure 29:

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```
CAL 3
CAL OK
[C0] Torque:RUN
[C1] 0.00 Nm = 1571
[C2] 135.58 Nm = 532364
```

Figure 29 – Command "CAL" Read Example

After the client requests channel calibration information for channel 3 ("Torque"), the DC-300 confirms a valid request, returns the *Calibration Status* and then sends any *Calibration Points* that are stored for this channel.

Perform Channel Calibration

To write new calibration points stored for a channel, the sequence of messages sent and received is illustrated in Figure 30:

```
1  CAL 3 START
   CAL OK
   [C0] Torque:CAL
2  CAL 3 ACCEPT 0
   CAL OK
   [C0] Torque:CAL
   [C1] 0.00 Nm = 6426
3  CAL 3 ACCEPT 50
   CAL OK
   [C0] Torque:CAL
   [C1] 0.00 Nm = 6426
   [C2] 50.00 Nm = 699115
4  CAL 3 END
   CAL OK
   [C0] Torque:RUN
   [C1] 0.00 Nm = 6426
   [C2] 50.00 Nm = 699115
```

Figure 30 - Command 'CAL' Write Example

The client first issues a request to begin calibration of the "Torque" channel **[1]**, and the DC-300 confirms the request and returns the *Calibration Status* which should now indicate **CAL**.

The client then adjusts the load on the dynamometer to zero and issues a request to accept the first calibration point as 0Nm **[2]**. The DC-300 confirms the request, sends the *Calibration Status*, and sends the first accepted calibration point.

The client then adjusts the load on the dynamometer to the max span and issues a request to accept the second calibration point as 50Nm **[3]**. The DC-300 confirms the request, sends the *Calibration Status*, and sends the first 2 accepted calibration points.

The client has finished the calibration and so issues a request to end the procedure **[4]**. The DC-300 confirms the request, sends the *Calibration Status* (which now specifies the channel is in **RUN** mode and not CAL mode), and then sends the now applied *Calibration Points*. The channel value will update to reflect the new calibration.

Abort Calibration

At any time during a calibration, the procedure can be aborted as shown in Figure 31:

```
CAL 3 ABORT
CAL OK
[C0] Torque:RUN
[C1] 0.00 Nm = 1571
[C2] 135.58 Nm = 532364
```

Figure 31 - Command 'CAL' Abort Example

The DC-300 confirms the request, sends the *Calibration Status* (which now specifies the channel is in **RUN** mode and not CAL mode), and then sends the original restored *Calibration Points* from before the calibration procedure began

Clear Calibration

Although rarely used, it is possible to remove all calibration points associated with a channel as shown in Figure 32:

```
CAL 3 CLEAR
CAL OK
[C0] Torque:RUN
```

Figure 32 - Command 'CAL' Clear Example

The DC-300 confirms the request and sends the *Calibration Status*. As all *Calibration Points* have been cleared, none will be sent.

4.1.1.13 Command: DC200 View (DC200)

The DC-300 is a replacement for an older display console called the DC-200. To allow channel engineering values sent from the DC-300 to be readable by a DC-200 serial client, it is possible to have the data formatted with the DC-200 protocol.

Client must send **<Command ID>** as '**DC200** with no parameters.

No custom response will be returned, however all subsequent "Channel Engineering Values" messages will now have the below format:

<Val "Speed"><TAB><Val "Torque"><TAB><Val "Power"><TAB><IMP or **MET>**

Where "Speed", "Torque", and "Power" engineering values are transmitted separated by a tab character. **<MET>** or **<IMP>** are appended depending on whether the channel units are metric or imperial respectively. Note that the DC-200 does not support additional temperature channels and so these are not included.

```
DC200
DC200 OK
0  28.44    0.00    MET
```

Figure 33 - Command 'DC200' Example

4.1.2 DCX Client

"DCX Client" is a Froude program developed for communicating with one or more consoles using the serial protocol. "DCX Client" presents channel data in a clean visual interface (Figure 34), and adds logging, plotting (Figure 35), and DC-200 compatibility as features not present in third party terminal emulation programs.

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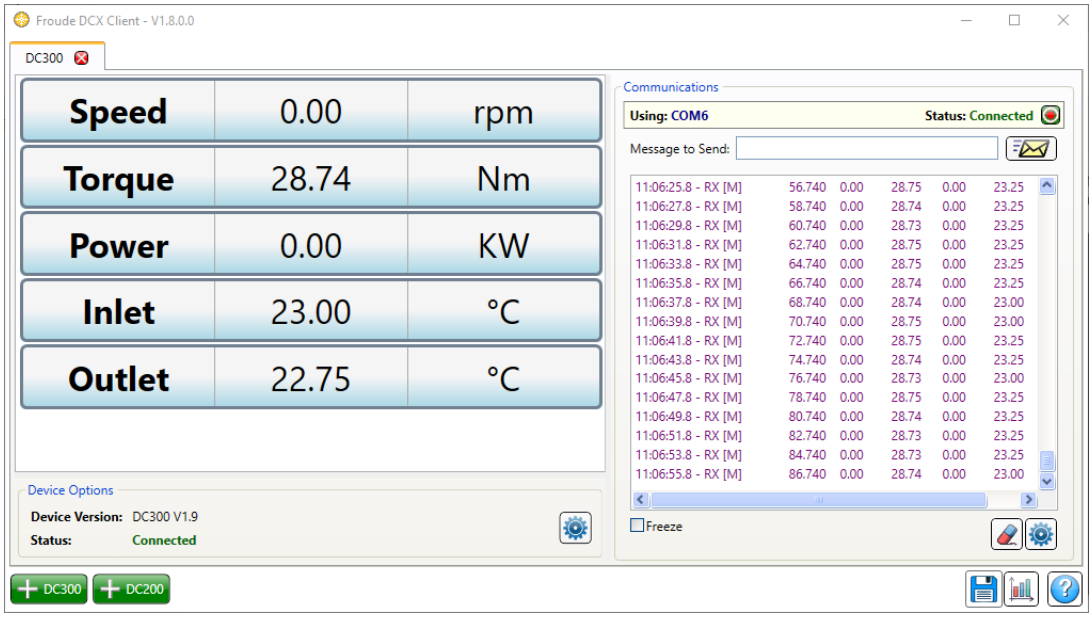

Figure 34 - DCX Client Gauges

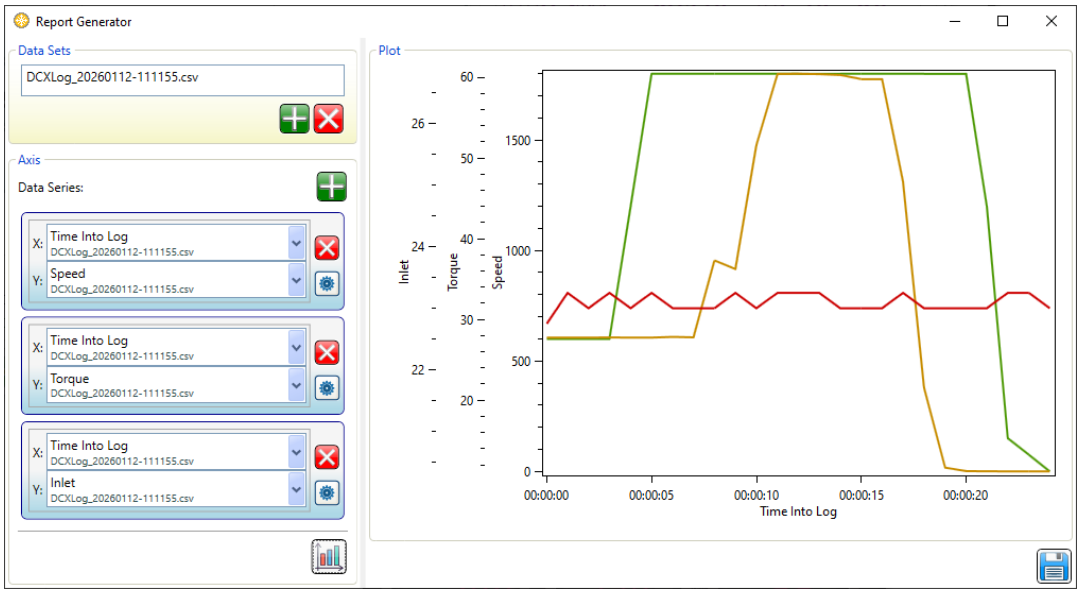

Figure 35 - DCX Client Plotting

Operation of “DCX Client” is provided in a manual included with the program, and so is not described here.

The “settings” button under “Device Options” is specific to the device type, and so the device specific configuration options for the DC-300 is shown in Figure 36.

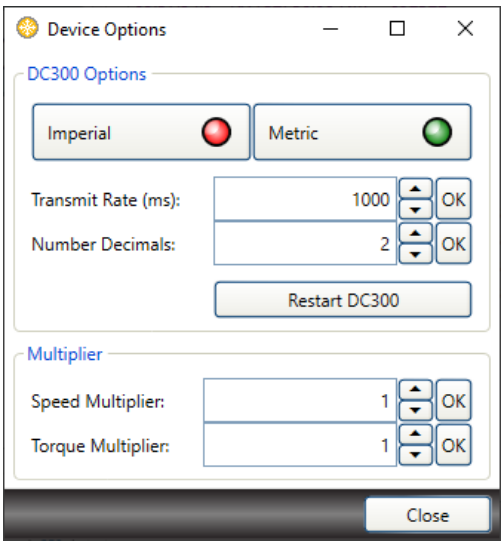

Figure 36 - DCX Client DC-300 Config

1 The “Imperial” and “Metric” buttons **[1]** trigger the IMP and MET commands. The active unit type with show with a green indicator.

2 The “Transmit Rate” **[2]** OK button sends the ENG command with defined millisecond broadcast rate.

3 The “Number Decimals” **[3]** OK button sends the DEC command with defined number of decimal places.

4 The “Restart DC300” **[4]** button sends the RST command and restarts the DC-300.

5 “Speed” and “Torque” multipliers are useful if using a gearbox- the on-screen channel values will be a product of the DC-300 transmitted value multiplied by these entered float values.

6

4.2 Wireless Host

Channel data can be accessed wirelessly on any device that has BLE (Bluetooth Low Energy)- this includes laptops, tablets, and mobile phones. This functionality is independent, which means one operator can be using the DC-300 console, another can be using the serial USB host, and a third could be walking around the test cell viewing channel data on their mobile phone. Only a single wireless host is currently recommended at a time.

This feature can be tested at https://froudedyno.github.io/DC300/

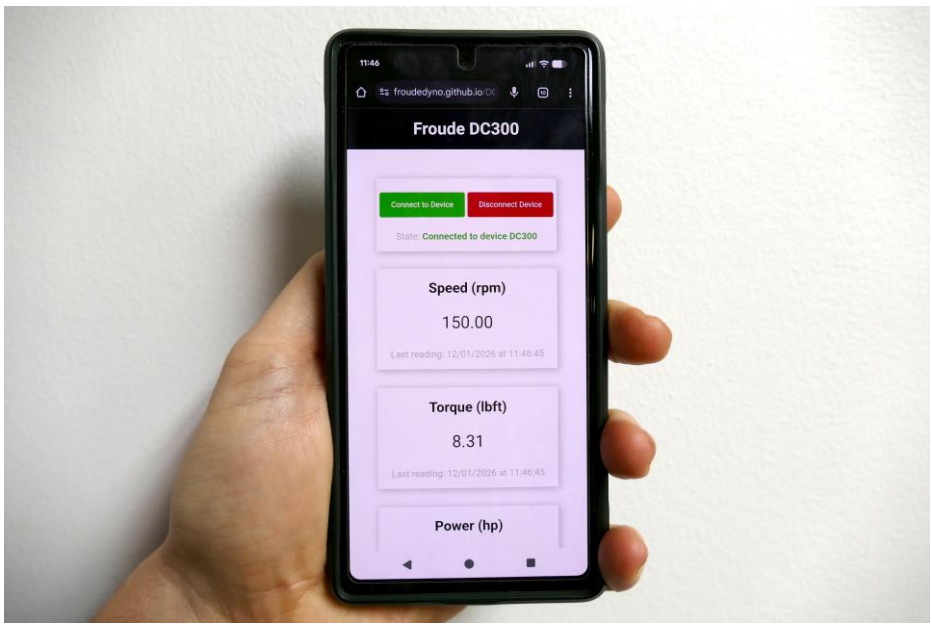

Figure 37 - Wireless Host Interface

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APPENDIX A CAL SHEET

Project Number:		**Ref:**	
Calibrated By:		**Date:**	
Calibration Units:		**Dynamometer:**	

Applied	Calculated	Measured	Notes
0	0		
ARM			